DISORDERS OF DNA REPAIR

Xeroderma Pigmentosum

Xeroderma pigmentosum (XP) serves as the prototype heritable disease with increased sensitivity to cellular injury. It is an autosomal recessive disorder characterized by sun sensitivity, photophobia, early onset of freckling, and subsequent neoplastic changes on sun-exposed skin. Patients exhibit cellular hypersensitivity to ultraviolet (UV) radiation and certain chemicals, due to defective DNA repair. Some individuals also develop progressive neurologic degeneration.

Frequency

XP occurs with an estimated frequency of 1 in 1 million persons in the United States and 1 per 100,000 people in Japan. The disease is more common in regions where consanguineous marriages are prevalent, such as the Middle East. Cases have been reported worldwide across all racial groups, including whites, Asians, blacks, and Native Americans.

▲ FIGURE 140-2 Xeroderma pigmentosum.
A. Pigmentary changes, atrophy, dryness, and cheilitis in a 16-year-old patient.
B. Cheek of a 14-year-old patient with pigmented macules of varying size and intensity, actinic keratosis, basal cell carcinoma, and a surgical scar.
C. Corneal clouding, prominent conjunctival blood vasculature, and loss of lashes.
D. Myriads of pigmented macules of varying size and intensity and scattered achromic areas on the back, with marked sparing of sun-protected buttocks of a 14-year-old patient.

Clinical Features

Skin

Approximately half of XP patients experience acute sunburn reactions following minimal UV exposure, while the remainder tan normally without excessive burning. In all patients, numerous freckle-like hyperpigmented macules develop predominantly on sun-exposed skin (see Fig. 140-2). Non-melanoma skin cancer typically develops at a median age of 8 years in XP patients—a 50-year reduction compared to the general population—highlighting the critical role of DNA repair in protecting against skin cancer. Multiple primary cutaneous neoplasms, including melanomas, are common.

Ocular Manifestations

Ocular abnormalities are nearly as prevalent as cutaneous findings and represent a key feature of XP (see Fig. 140-2C). The retina is shielded from UV radiation by anterior structures (eyelids, cornea, and conjunctiva), so clinical involvement is largely confined to these exposed areas. Photophobia is often an early symptom.

Internal Malignancies

Review of global XP literature reveals an increased incidence of internal neoplasms. Oral cavity cancers—particularly squamous cell carcinoma of the tongue tip, a sun-exposed site—are frequently reported. Rare cases include brain tumors (e.g., sarcoma, medulloblastoma), spinal cord astrocytomas, and cancers of the lung, uterus, breast, pancreas, stomach, kidney, and testis, as well as leukemia. Overall, XP patients have an estimated 10- to 20-fold increased risk of developing internal malignancies.